

Name:

Collaborators:

Instructions: Worksheets are graded mostly on completion and partially on correctness. Please write complete solutions showing explanations and key steps to the following problems, unless it says otherwise.

Find Eigensolutions with Complex Eigenvalues

1. Imaginary Eigenvalues

Consider the *linear homogeneous system of 1st-order ODEs with constant coefficients*:

$$\begin{aligned} x' &= y \\ y' &= -2x \end{aligned}$$

- a. Show that the eigenvalues are purely imaginary

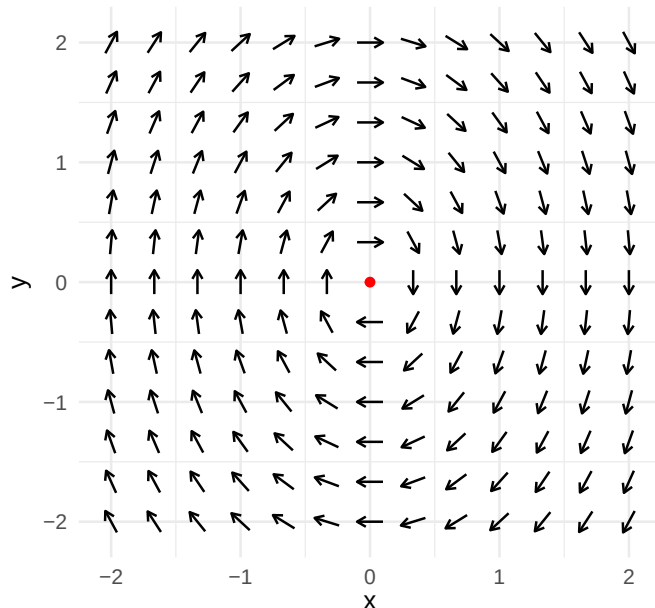
$$\lambda_1 = i\sqrt{2} \quad \text{and} \quad \lambda_2 = -i\sqrt{2}.$$

- b. Show that the corresponding eigenvectors are

$$\vec{v}_1 = \begin{bmatrix} -\frac{1}{\sqrt{2}}i \\ 1 \end{bmatrix} \quad \text{and} \quad \vec{v}_2 = \begin{bmatrix} \frac{1}{\sqrt{2}}i \\ 1 \end{bmatrix}.$$

Then, write these eigenvectors as a linear combination of its real and imaginary parts.

- c. Below is the vector field for the given system. Draw the real and imaginary parts of the eigenvectors on the phase plane.



- d. Draw a specific solution curve on the direction field with initial condition $(x, y) = (1, 0)$. Then, discuss how the parts of the complex eigenvectors help determine the orientation of the ellipse formed by the solution.

2. Complex Eigenvalues

Consider the *linear homogeneous system of 1st-order ODEs with constant coefficients*:

$$\begin{aligned} x' &= y \\ y' &= -2x - 2y \end{aligned}$$

- a. Show that the eigenvalues are complex conjugates

$$\lambda_1 = -1 + i \quad \text{and} \quad \lambda_2 = -1 - i.$$

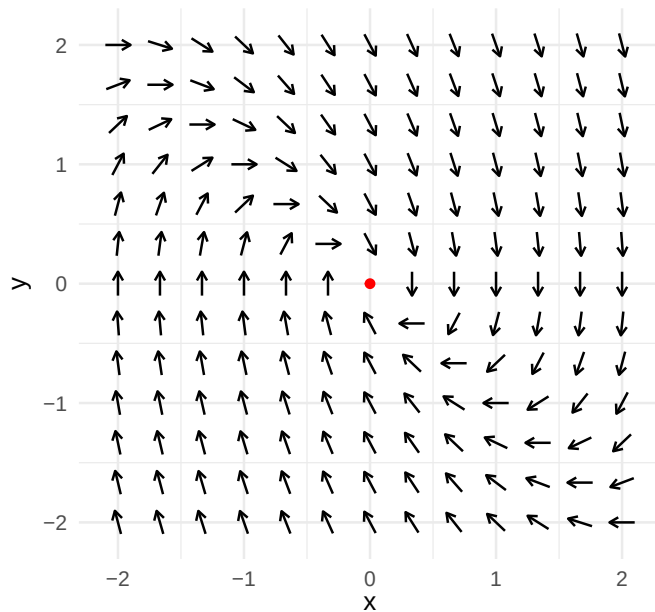
- b. Show that the corresponding eigenvectors are

$$\vec{v}_1 = \begin{bmatrix} -1 - i \\ 2 \end{bmatrix} \quad \text{and} \quad \vec{v}_2 = \begin{bmatrix} -1 + i \\ 2 \end{bmatrix}.$$

Then, write these eigenvectors as a linear combination of its real and imaginary components.

- c. Determine the general solution. Here, you can use λ_1 and $\vec{v}_1$ to form a solution using Euler's formula. Then, use the real and imaginary parts of this solution as two linearly independent solutions to form a general solution.

- d. Below is the vector field for the given system. Draw the real and imaginary parts of the eigenvectors on the phase plane.



- e. Draw a specific solution curve on the direction field with initial condition $(x, y) = (1, 0)$. Then, discuss how the eigenvectors help determine the orientation of the spiral formed by the solution.
- f. Determine the specific solution given the initial condition $(x, y) = (1, 0)$.